

GEOTECHNICAL INVESTIGATION REPORT

Project: Sunrise Residential Complex

Date: May 16, 2026

1. INTRODUCTION

This report presents the geotechnical data for the proposed building site.

2. BOREHOLE LOG (BH-1)

Depth 0.0m to 1.5m: Silty Clay (CI), N = 12. Description: Brown firm silty clay.

Depth 1.5m to 4.0m: Sandy Silt (SM), N = 24. Description: Yellowish dense sandy silt.

Depth 4.0m to 8.0m: Weathered Rock, N > 50. Description: Highly weathered granite.

3. GROUNDWATER

Water table encountered at 3.5m depth during investigation.

4. BEARING CAPACITY

Calculated Safe Bearing Capacity (SBC) for shallow foundation: 250 kPa at 2.0m depth.

Pile capacity (300mm dia bored pile): 450 kN at 8.0m depth.

5. RECOMMENDATIONS

Based on the soil profile, isolated spread footings are recommended at 2.0m depth.

6. RISKS

Risk of liquefaction in the upper sandy silt layer during seismic events.

7. IS CODES

Analysis done as per IS 1892 and IS 6403.